GraphMind: Efficiently Exploring Multi-Agent LLM Topologies with Graph Neural Networks

Vito Levstik and Gal Zmazek

University of Ljubljana

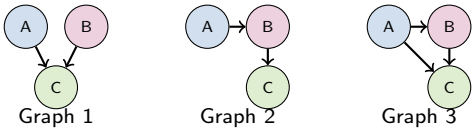
January 11, 2026

The Challenge: Topology Search is Expensive

The Problem:

- Multi-agent LLM systems need configuration
- Which agents? How to connect them?
- Combinatorial explosion: millions of possible graphs
- Each evaluation requires many LLM API calls
- **Cost: $$$ + Time**

Our Solution:
Use GNNs as surrogate models to predict performance



. . . millions more . . .

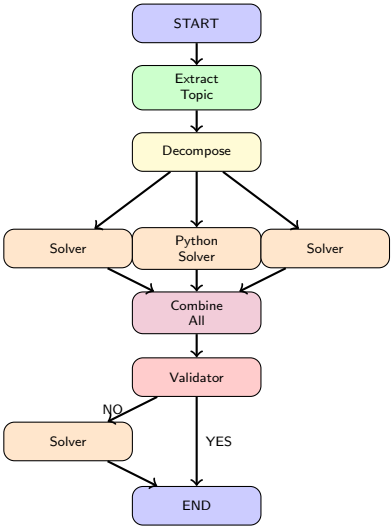
Which one is best?

Multi-Agent System for Mathematical Problem Solving

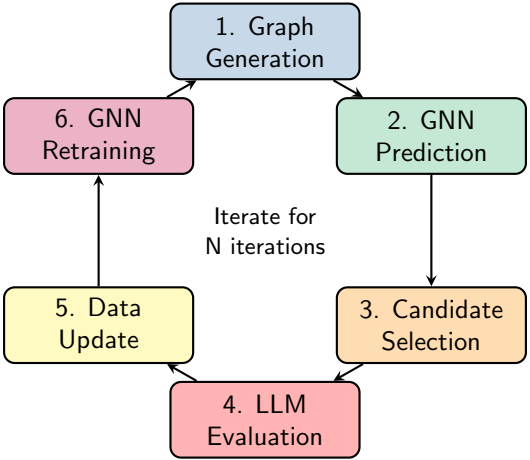
8 Specialized Agent Types:

- Solver: Generate solutions
- Python_solver: Execute code
- Validator: Check correctness
- Extract_topic: Identify concepts
- Decompose: Break into subproblems
- Split: Parallel exploration
- Combine_all: Merge results
- Explain: Provide reasoning

Task: Solve math problems from NVIDIA OpenMathInstruct-1 dataset



GraphMind Pipeline: 6-Step Iterative Process

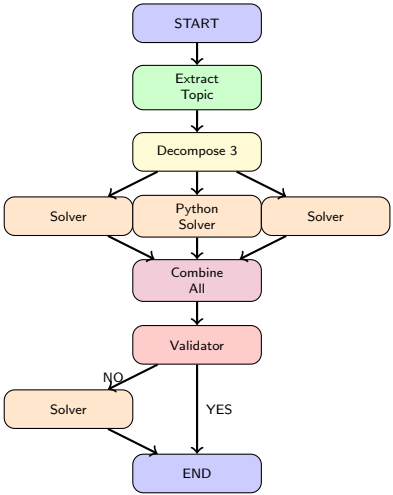


Key Idea: Generate many candidates → GNN filters → Evaluate only the best → Improve GNN

Step 1: Graph Generation

Semi-Random Generation:

- Generate 200,000 graphs per iteration
- Constraints: max 8 nodes, max depth 3
- Rules enforce structure:
  - Decompose_3 $\rightarrow$ 3 branches
  - Combine_all merges branches
  - START and END nodes
- Filter duplicates
- Result: 1,000-2,000 unique graphs



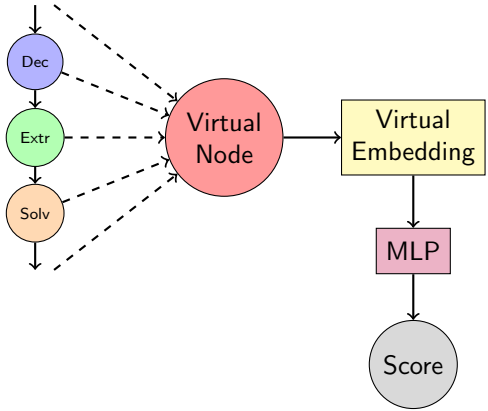
Step 2: GNN Prediction with Virtual Node

GNN Architecture:

- **Input:** One-hot node features (15 types)
- **Models:** GCN, GraphSAGE, GAT
- **Virtual Node:** Connected to all nodes
- **Message Passing:** 2-4 layers
- **Output:** Score $\in$ [0,1] via sigmoid

Virtual Node Technique:

- Aggregates global graph information
- Final embedding $\rightarrow$ MLP $\rightarrow$ prediction
- Enables graph-level predictions



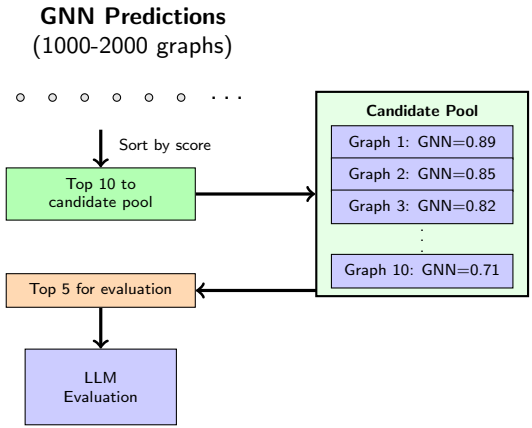
Steps 3-4: Selection and LLM Evaluation

Step 3: Candidate Selection

- Top 10 graphs $\rightarrow$ candidate pool
- Select top 5 for evaluation
- Pool size limited to 10
- Removes lowest-scoring graphs

Step 4: LLM Evaluation

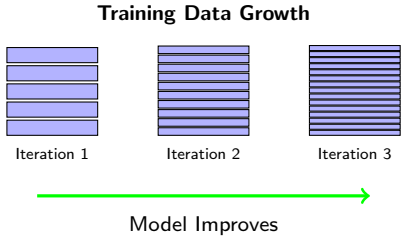
- Build LangGraph from structure
- Evaluate on 5 math problems
- Each agent makes LLM calls
- Thousands of API calls!
- Score: exact match or relative error



Steps 5-6: Data Update and GNN Retraining

Step 5: Data Update

- Add evaluated graphs to training set
- Each graph: structure + true score
- Dataset grows by 5 graphs/iteration
- Accumulates over iterations



Step 6: GNN Retraining

- Retrain every iteration
- Supervised learning: MSE loss
- Adam optimizer
- 300 epochs, batch size 32
- Model learns structure-performance mapping

Random Baseline: Understanding the Distribution

Establishing a Baseline:

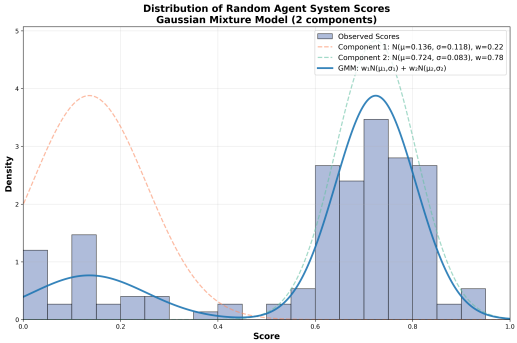
- Generated 5M random graphs
- Sampled 150 for evaluation
- Measured performance on math problems
- Analyzed score distribution

Observed: Bimodal Distribution

- Two clusters: good (> 0.5) and poor (< 0.3)
- Simple distributions insufficient

Solution: Gaussian Mixture Model

- GMM with 2 components
- Fitted using EM algorithm
- Captures bimodal structure
- Baseline for comparison



Random graph distribution fitted with GMM
Two clusters: good and poor performing graphs

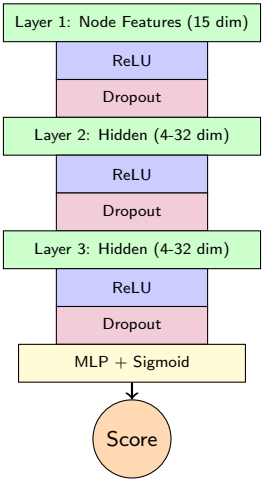
GNN Models: Architecture Details

Three GNN Architectures:

1. **GCN** (Graph Convolutional Network)
2. **GraphSAGE**
3. **GAT** (Graph Attention Network)

Hyperparameter Search:

- Layers: [2, 3, 4]
- Hidden dim: [4, 8, 16, 32]
- Dropout: [0.0, 0.1, 0.2, 0.5]



Results: GNN Model Performance

Hyperparameter Search Results:

Model	Layers	Hidden	MSE
GCN	4	8	0.0123
GraphSAGE	4	16	0.0089
GAT	4	8	0.0156

Test Performance (10 runs):

Model	Test MSE
GCN	0.0145 ± 0.0021
GraphSAGE	$\mathbf{0.0102 \pm 0.0018}$
GAT	0.0178 ± 0.0025

GraphSAGE performs best!

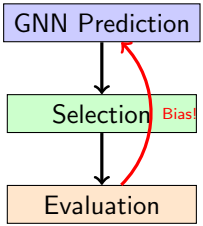
Key Insights:

- 4 layers optimal
- Hidden dim 8-16 works well
- GraphSAGE generalizes best
- Low variance across runs

Pipeline Dynamics: The Prediction Paradox (1/2)

An Interesting Phenomenon:

- GNN predictions become very accurate
- But... this creates a bias!
- Model overfits to its own selections
- Training data becomes skewed



Bias Loop: Model selects what it knows → learns only those patterns

The Feedback Loop:

1. GNN predicts good graphs
2. Only predicted "good" graphs evaluated
3. Training data biased toward these
4. GNN learns to predict "more of same"
5. Diversity decreases over iterations

Result: Training data becomes increasingly biased toward model's preferred patterns

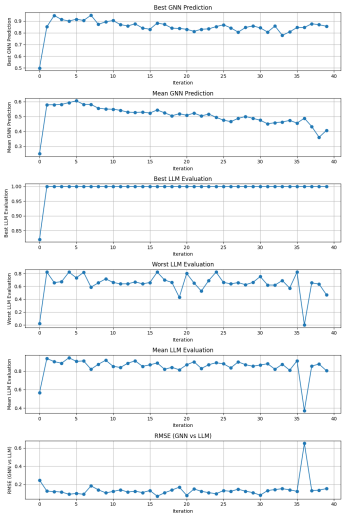
Pipeline Dynamics: The Prediction Paradox (2/2)

Observable in Metrics:

- RMSE appears to improve
- But on biased sample!
- GNN score variance increases
- Actual diversity decreases

Key Insight:

- Model becomes "too good" at its own strategy
- Reduces exploration of search space
- Selection bias accumulates



Results: GNN-Guided vs Random Search

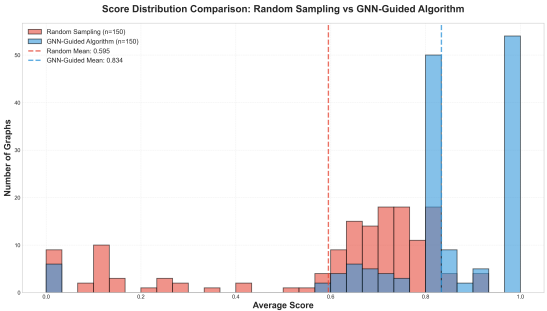
Experimental Setup:

- **Baseline:** 150 random graphs
- **GraphMind:** 30 iterations
- Each graph evaluated on 10 problems
- Modeled random baseline as GMM

Key Results:

- GraphSAGE: Best average (0.72 vs 0.43)
- GAT: 10% of graphs scored 1.0
- GCN: Strong performance (0.69 avg)
- All models significantly beat random

Score Distributions



Random baseline (red) vs GNN-guided search (blue)

Challenges and Future Work

Current Limitations:

- Selection bias feedback loop
- Limited to math problems
- High variance in LLM evaluations
- Fixed evaluation budget

Future Directions:

1. **Mitigate Bias:**
   - Stratified sampling
   - Diversity-aware selection
   - Exploration-exploitation balance
2. **Smart Generation:**
   - Learn from failures
   - Exclude similar bad graphs
3. **Progressive Evaluation:**
   - Start with fewer problems
   - Full eval only for promising graphs

Conclusion

Key Takeaways:

- GNNs can effectively predict multi-agent system performance
- Iterative refinement improves predictions over time
- Massive cost savings enable practical topology search
- GraphSAGE architecture works best for this task

GitHub: `github.com/vitolev/GraphMind`

Thank you! Questions?